



From Arc to Filamentary Plasma: Airflow-Controlled Regime Transition in Atmospheric Pressure DC Gliding Arc Discharge

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Abstract

Gliding arc discharge (GAD) plasmas operated in atmospheric air exhibit complex transitions between thermal arc and non-thermal filamentary modes, which significantly affect their physicochemical properties. In this study, the effect of airflow on the plasma regime transition in atmospheric direct current (DC) gliding arc discharge was experimentally investigated. The plasma was generated using copper wire electrodes with a fixed 6 mm spacing, and the airflow rate was controlled by varying the voltage of a DC snail fan. Electrical properties were analyzed using a high-voltage probe, current measurements, and oscilloscope diagnostics, while plasma emission was studied using optical emission spectroscopy. The results reveal a clear transition from a stable arc regime at low airflow to a non-thermal filamentary regime at higher airflow rates. The optical emission spectra are dominated by a nitrogen-second positive system with strong emission at 337 nm and weaker first negative system bands around 391 nm. The electron temperature was determined using the intensity ratios of these emissions. The vibrational temperature was determined using a Boltzmann plot method based on N₂ emission bands at 337, 357, and 380 nm. Additionally, the reduced electric field (E/N) decreased with increasing airflow due to simultaneous decreases in discharge voltage and gas temperature variations. A plasma regime map relating airflow velocity and discharge voltage was constructed, revealing three distinct regions: arc, transition, and filamentous regimes. These findings demonstrate that airflow plays a crucial role in controlling the electrical properties, excitation dynamics, and discharge structure of atmospheric DC GAD plasmas.

Keywords Gliding arc discharge · Plasma regime transition · Characterization · Diagnostics · Optical emission spectroscopy

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Introduction

Atmospheric pressure plasma technologies have attracted significant attention in the last two decades due to their applications in environmental remediation, plasma-assisted combustion, gas conversion, and biomedical sterilization [1–4]. This interest stems from the fact that plasma sources capable of operating under atmospheric conditions are less expensive, more scalable, and easier to integrate into industrial applications compared to vacuum systems. Among the various atmospheric plasma sources, gliding arc discharge (GAD) systems stand out as a unique hybrid plasma configuration that incorporates both the high reactivity characteristics of thermal arc discharges and the low gas temperature advantages of non-thermal plasmas [5, 6].

A gliding arc discharge is typically generated between two electrodes that are diverging (far apart). Plasma ignites at the narrowest point of the electrodes and then extends along them, driven by the flow of gas injected into the system [1, 7]. This formation mechanism is based on the principle of the Jacobs ladder [8]. This dynamic behavior of the discharge results in a non-thermally equilibrium plasma characterized by a high electron temperature (T_e) but a relatively low gas temperature (T_g) [9, 10]. This geometry and flow-forced elongation move GAD into a hybrid regime that distinguishes it from both classical thermal arcs and typical cold (non-thermal) discharges. Therefore, GAD is often positioned in the literature as a “warm plasma” where near-equilibrium (thermal) and non-equilibrium (non-equilibrium) properties can be observed together [6, 11]. This “warm” character provides a high reactivity potential thanks to the energies of the electrons in the eV range, while simultaneously creating a plasma-heat interaction that significantly increases the gas temperature [12, 13]. The spatiotemporal evolution of GAD is generally described in modern studies using “breakdown–elongation–extinction” cycles [14, 15]. This cycle can exhibit regime transitions between stable gliding mode and unstable breakdown mode depending on the applied power, voltage level, and gas flow rate [16, 17]. In the stable mode, the arc persists for a longer period, while in the unstable mode, short-lived but periodic re-ignitions are observed, characterized by a highly reduced electric field and higher electron temperature.

Therefore, the combined evaluation of simultaneous electrical waveforms, fast imaging, and optical emission spectroscopy (OES) measurements is highlighted in the literature for characterizing GAD [18–21].

The electrical and physical characteristics of the discharge depend heavily on various parameters, including the applied voltage, gas composition, electrode geometry, and gas flow rate [16, 22, 23]. Among these parameters, gas flow rate, arc elongation, discharge stability, and plasma residence time within the reactor play a particularly important role in determination [24, 25]. Gas flow dynamics significantly influence arc propagation and plasma chemistry [12, 26]. Increasing airflow velocity can enhance arc elongation and expand the interaction volume between plasma species and the surrounding gas [27, 28].

For example, a positive correlation has been reported between jet length and airflow velocity in atmospheric pressure gliding arc plasma jet (GAPJ) systems [29]. In studies involving rotating gliding arcs, it has been shown that increasing input voltage or decreasing gas flow rates support the stable gliding mode [16, 30]. However, excessive airflow can reduce the residence time of reactive species and lead to partial extinction of the plasma discharge [23, 24]. This demonstrates that flux is not merely a parameter that “extends the

arc,” but also shapes electrical and optical oscillations through events such as re-ignition frequency, arc root bounce, and refraction [15, 31].

OES is a widely used, non-invasive diagnostic technique for characterizing atmospheric plasma systems [32–34]. In gliding arc discharges generated using air, the emission spectrum is dominated by nitrogen molecular bands belonging to the second positive system (SPS) of N_2 and the first negative system (FNS) of N_2^+ [27, 35, 36].

The emission band at 337 nm corresponds to the N_2 ($C^3\Pi_u \rightarrow B^3\Pi_g$) transition, while the emission band at 391 nm corresponds to the N_2^+ ($B^2\Sigma_u^+ \rightarrow X^2\Sigma_g^+$) transition [37]. These spectral lines and bands are frequently used to estimate plasma parameters such as electron temperature, vibrational temperature (T_{vib}), and rotational temperature (T_{rot}) [38, 39]. In atmospheric pressure air plasmas, it is widely accepted that the T_{rot} temperature converges to the gas temperature (T_g) [34, 40]. The ratio of N_2 and N_2^+ emission densities (specifically the 391 nm/337 nm ratio) provides valuable information about electron energy distribution and is used as a practical method for electron temperature estimation in atmospheric air plasmas [41, 42].

In air gliding systems, rotational and vibrational temperature extraction from OES is a very common practice in the current literature and is usually implemented by matching N_2 band systems with theoretical spectra (e.g., with Specair or Lifbase software) [34, 40, 43]. For example, a GAPJ study showed that rotational temperatures were reported to be in the range of ~ 1373 – 2065 K and vibrational temperatures in the range of ~ 2405 – 2700 K under different operating conditions [27]. In the rotating gliding arc, T_{rot} and T_{vib} measurements for N_2 varied depending on the regime; for example, T_{rot} was reported at ~ 2200 – 2600 K and T_{vib} at ~ 4400 – 4820 K [30]. In high-speed flows with low-current gliding DC discharge, much higher values of T_g (rotational) of ~ 7000 – 9500 K and T_v of ~ 7000 – $11,000$ K have been reported, and the differences in radiation radii along the cross-section, as well as the radial temperature gradient and composition homogeneity, have been discussed [18, 44]. When these studies are examined together, two critical conclusions emerge: (i) Air gliding discharges can span a wide thermal regime band; even within the same gliding topology, temperatures can vary on a scale of thousands of Kelvin as a result of flow velocity, current/voltage, and geometry [13, 45]. (ii) While the T_{vib} – T_{rot} dissociation (out-of-equilibrium) is maintained under some conditions, under others a tendency towards equilibrium can be observed due to high T_g ; this point is one of the main reasons why GAD is referred to as “warm plasma” [9, 11].

In air gliding systems, “electron temperature” is reported in different ways in the literature. Some studies show the electron excitation temperature (T_{exc}) using a Boltzmann plot [46, 47]. Some studies report regime-dependent average T_e values in the eV range; for example, in a rotating gliding arc study, values of approximately 1.90 eV were obtained in the unstable breakdown mode, and approximately 0.64 eV in the stable gliding mode [16]. In model-assisted GAD reforming studies, characterization is performed using a representative value of approximately ~ 1.5 eV for T_e [10, 48]. The GAPJ study also shows that T_{exc} varies within a narrow range (approximately 1.36–1.44 eV) [29]. This variation is important because the “ T_e ” derived from OES may not always be the true thermodynamic electron temperature based directly on the electron energy distribution function (EEDF), and should often be interpreted as the excitation temperature [34, 40]. Indeed, the literature emphasizes that the excitation temperature obtained with OES can be differentiated from the T_e mea-

sured by Thomson scattering, and that the deviation from LTE can be observed, making it a prominent diagnostic approach [42].

In GAD and DC arc-like discharges, the cathode region is crucial for discharge stability and voltage-current characteristics [49]. In the context of GAD, it is argued that in low and medium current regimes, electron emission can shift to thermionic emission as the cathode temperature increases, while in more thermal arc-like regimes, field emission can become dominant due to cathode erosion and microstructure changes [50, 51].

Similarly, it has been reported that in glow-like DC air discharges, secondary electron emission can be altered by oxidation of the cathode material, which can reduce energy consumption via the cathode drop-off voltage; ultimately improving energy costs without significantly changing the nitrous oxide production rate [52, 53]. In this context, the use of DC power supplies stands out compared to AC systems due to its ease of installation, controllability, and the ability to more directly examine cathode physics processes [48, 54]. Current literature on experimental studies of DC-driven gliding discharges in air shows that gliding DC discharges can be sustained even at high-speed airflows within certain current ranges (up to the order of a few amperes) [55, 56]. In these types of studies, discharge is initiated with a high-voltage DC source (kV range) and the resulting species are monitored with OES [45, 57].

In addition to plasma diagnostics, many studies have also investigated the thermal behavior of gliding arc systems [21, 44, 49]. Although classified as non-thermal plasmas, localized heating can occur within the arc column due to Joule heating and ion bombardment [50, 58]. In addition to OES-based rotational temperature measurements for determining gas temperature, there are also studies correlating IR thermography and optical fiber-based temperature measurements with OES to understand surface/electrode temperature and thermal effects [21, 59]. For example, in air DBD and air gliding arc comparisons, the combined interpretation of IR/fiber measurements with the temperature approximation via OES was used to establish the thermal effect-surface functionalization relationship [4, 60]. Alternatively, measuring gas temperature by positioning an electrically insulating thermocouple at a specific distance also stands out as a practical method [47].

The motivations for applying GAD are discussed, particularly in processes such as nitrogen oxidation, nitrogen fixation, CO₂ conversion, pollutant removal, and plasma-assisted combustion, based on its ability to operate at atmospheric pressure, its wide range of interactions, its modular scalability, and its ease of implementation [5, 6, 61–67]. For example, in the context of scramjet combustion, it has been stated that the gliding arc offers a wider area effect and is an easily achievable method [9]. However, the same literature also highlights that in GAD, due to the high gas temperature, vibrational energy accumulation (T_{vib}) can rapidly approach T_g via translational relaxation, and this can be limiting in terms of chemical efficiency and energy cost [13, 50]. From an energy cost perspective, the literature also shows that GAD is not always the most efficient option: for example, magnetic field-stabilized atmospheric air glow discharge has been reported to significantly reduce the energy cost of NO_x production compared to GAD, and offers the advantage of having all the gas pass through the plasma region [26]. It has been reported that N₂, N₂⁺, NO, OH, and atomic O/N lines are prominent in the OES spectra of air gliding arc discharges; and that the intensities of the lines and bands associated with N₂⁺, NO, and O increase with increasing power and voltage [30, 68, 69]. In the rotating gliding arc study, it was argued that NO_γ band emission increased with regime, and that longer arc duration and higher gas

temperature might be associated with increased NO_γ intensity [30, 68]. Furthermore, it has been shown that temperature analysis can be performed via the second positive system of N_2 using OES in other atmospheric air discharges near GAD (such as DC glow), and that species distributions vary with electric field profiles [34, 55]. This situation methodologically strengthens the possibility that OES-based temperature derivations are applicable in GAD, at least in certain regions [35, 38].

Commonly used air sources include compressed air compressors, compressed dry air gas cylinders, and electric snail fans [29, 70]. In GAD studies, it is emphasized that optical and temperature measurements should be evaluated not in isolation, but together with V-I characteristics and discharge cycle; because ignition, shortcut events, power consumption, oscillations and arc evolution are manifested in these signals [15, 20, 71]. The rotating gliding arc study offers an exemplary methodology in this context: the electrical and optical projection of the breakdown-elongation-extinction phases in a cycle is established through the synchronous use of a high-speed camera, oscilloscope, and OES [16]. The GAPJ study also reported the combined effects of power, RMS current/voltage, and flow rate/voltage [27].

The literature summarized above shows that systematic research focusing on the effect of airflow velocity on plasma parameters in DC-operated gliding arc reactors is relatively limited [23, 68]. Current studies generally examine specific parameters in a particular configuration, but do not offer a comprehensive approach that reveals the simultaneous and holistic effect of flow rate on electron temperature, vibration temperature, gas temperature, arc length, cathode temperature, and reactive species formation. Therefore, our current study aims to systematically investigate the effect of airflow velocity on electron temperature, vibration temperature, arc length, cathode temperature, and gas temperature in a DC gliding arc discharge system using OES. Simultaneously, the changes in the emission intensities of radical nitrogen and oxygen species (RONS) (such as N_2 , N_2^+ , NO , OH , atomic O , and N) formed during discharge are also analyzed, and the effect of airflow velocity on plasma chemistry is explored. The findings of this study will contribute to the optimization of GAD reactors and the determination of suitable operating conditions for various applications.

Materials and Methods

In the experimental study, a transparent plexiglass casing was constructed to set up a gliding arc discharge reactor. A 3 mm diameter copper wire was used as the electrode. The copper wires were bent to form a divergence geometry to create the electrode structure for the gliding arc discharge. The closest distance between the electrodes was set at 6 mm. This distance parameter was chosen to facilitate the initiation of the discharge and to prevent the discharge from extinguishing under high airflow conditions. Therefore, the distance was kept constant throughout the experiment. A DC high-voltage flyback transformer and a zero-voltage switching (ZVS) driver system with Mazzilli circuit topology were used as the power source to generate the discharge [54]. The power supply was manufactured in our laboratory using our own resources. The design ensures a constant output voltage. Using a 1000X attenuation high-voltage probe, we measured the output of the power supply and observed a value of 22 kVrms under no load. The output current was measured as 36 mA directly during the arc. To measure the current, we added a 100-ohm, 1 W metal film resistor between the negative electrode and the ground line. Thus, using Ohm's law, we measured

the current in rms by calculating the voltage across the resistor. A DC motor-controlled snail fan blowing atmospheric air was used as the discharge gas source. The snail fan's motor operates between 0 and 12 V. The airflow speed can be controlled by varying this voltage range. This approach provided a simple and cost-effective method for controlling the plasma operating regime. The airflow was injected from the bottom of the reactor and forced the discharge channel to move along the diverging electrode geometry, producing the characteristic gliding arc motion. When a high voltage was applied, the discharge initially ignited in the narrowest gap between the electrodes where the electric field strength was highest. Due to the aerodynamic force created by the airflow, the arc column moved downwards along the separated electrodes, increasing in length and eventually extinguishing. Then, a new arc was reignited at the base of the electrode, creating a periodic gliding arc discharge regime. Figure 1 shows the experimental schematic of the system.

To observe the spectral lines in detail during discharge formation, an ASEQ LR1 optical emission spectrometer with a measurement range of 190–1200 nm, a slit spacing of 50 μm , and a spectral resolution of 2 nm was used. The optical fiber was positioned approximately 5 cm from the discharge region and aligned perpendicular to the plasma column. A Rigol DHO804 digital oscilloscope was used to observe the output voltage and output current waveform. A K-type thermocouple and a CEM DT613 monitor were used to measure the gas temperature during plasma formation. The gas temperature was measured using a K-type thermocouple positioned approximately 1 cm from the electrode region. A Seek XR mobile thermal camera was used to measure the cathode temperature during the discharge. The airflow velocities of a snail fan were scaled using different operating voltages. A Uni-T UT363 anemometer was used for this purpose. Airflow velocities were determined by sequentially scanning the operating voltages at 1-2-3-4-5-6-7-8-9-10 V. Based on these voltage values, the airflow velocities were measured as 4-8-12-16-20-24-28-32-36-40 m/s, respectively.

Table 1 shows the numerical parameters used during the experiment.

Figure 2 shows the GAD reactor, the DC high-voltage power supply, and the voltage-current regulated DC power supply used to drive the snail fan. The voltage and current values of the snail fan are controlled by the DC power supply.

In Fig. 3, measurements were taken using a high-voltage probe with the output terminals of the DC high-voltage power supply unloaded. The V_{rms} value measured was approxi-

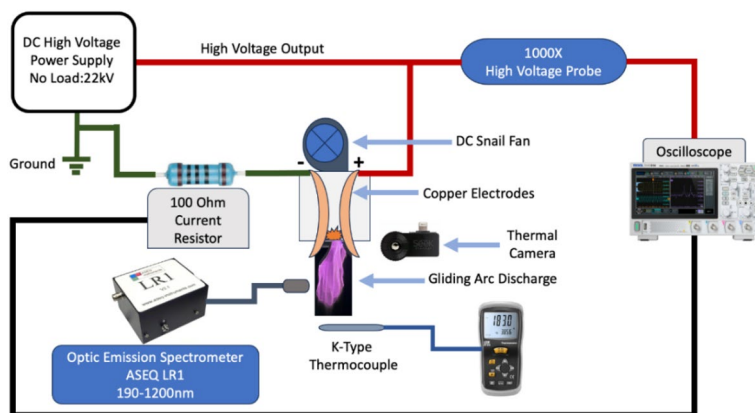


Fig. 1 Experimental setup

Table 1 Numerical parameters used as setup during the experiment

Parameter	Value
Electrode gap	6 mm
Air velocity	4–40 m/s
Voltage	(No load:22 kV) (Load:2–3 kV)
Spectral range	190–1200 nm
Fiber optic cable distance	5 cm
Thermocouple distance	1 cm

**Fig. 2** The components used in the experiment were: (a) the GAD reactor, (b) the DC high-voltage power supply, and (c) the DC power supply used for the snail fan**Fig. 3** Oscilloscope image of the power supply output voltage when there is no discharge

mately 22 kV. It was observed that the waveform on the oscilloscope screen was displayed as direct current.

OES graphs of discharges obtained at different airflow velocity values during the experiment were recorded. Gas temperatures were measured using a thermocouple. A silicone tube was attached to the probe body during thermocouple measurement. This was to prevent the digital thermocouple monitor from being electrically affected by the high electric field during discharge generation. This ensured accurate measurement of the gas temperature. Then, using the obtained OES lines, electron temperature and vibrational temperature values were

measured for different airflow velocities. To determine the electron temperature, 391 nm (N_2^+ first negative system) and 337 nm (N_2 second positive system) spectral lines were used. These spectral lines were used to estimate the electron temperature using the density ratio method.

In the literature, the intensity values of these lines are used. However, due to the low spectrometer resolution and the fact that the intensity values are in arbitrary units (a.u.), the area under the lines was calculated using the Gaussian fitting method. Therefore, areas were used instead of intensity values. The ratio 391/337 was determined for each airflow velocity. This method is widely used in atmospheric air plasma diagnosis because the excitation thresholds of these transitions are highly dependent on electron energy [41]. Electron temperature predictions were made using calibration curves from the literature for these values. The vibrational temperature was estimated using the Boltzmann plot method based on the N_2 second positive system emission bands at 337, 357, and 380 nm [72].

Statistical analysis was performed using the intensities of the spectra obtained from OES measurements. Since repeated measurements were not performed under experimental conditions, descriptive statistics and effect size analyses were preferred. To evaluate the effect of plasma regimes (arc, transition, filamentary) on spectral properties, the classification of regime transitions obtained depending on airflow velocity was used as the basis. For each plasma regime, descriptive statistics were presented by calculating the mean and standard deviation (SD) of the emission intensities of the selected principal spectral lines (N_2 337 nm, N_2^+ 391 nm, NO 235 nm, OH 309 nm, N_2 357 nm, and N_2 315 nm). To quantitatively express the practical significance of the differences observed in the transition from arc to filamentary regimes, the effect size coefficient Cohen's *d* was calculated. Cohen's *d* values were obtained by dividing the difference between the means of the groups by the pooled standard deviation and interpreted according to the threshold values in the literature (*d* < 0.2: insignificant, 0.2–0.5: small, 0.5–0.8: moderate, 0.8–1.3: large, > 1.3: very large effect) [73]. Then, depending on the different airflow rates, the gas density was calculated using the gas temperature data for each flow rate. Electric field values were determined using fixed electrode spacing and voltage values. Using this data, the electric field reduction ratio as a function of gas density was calculated [74]. The total power dissipated at different airflow velocities (DC high-voltage power supply + snail fan DC power supply) was analyzed. Thus, power consumption graphs were created and approximate efficiency was calculated.

Results and Discussions

Electrical Characteristics of the Discharge

The electrical characteristics of an atmospheric DC gliding arc discharge were investigated as a function of airflow velocity. The airflow was controlled by varying the supply voltage of the snail fan, and flow velocities in the range of approximately 4–40 m/s were obtained. Experimental results showed that the discharge voltage decreased significantly with increasing airflow velocity. The voltage measured at the lowest flow rate was approximately 3 kV, while at the maximum airflow rate, the voltage dropped to approximately 2 kV. This decrease can be attributed to the elongation of the plasma channel by the airflow and the simultaneous increase in the convective cooling effect. The discharge current also varied depending on the airflow velocity. The measured current values decreased gradually from

approximately 33.6 mA to 31.7 mA. Compared to the change in voltage, the change in current appears to be more limited. This indicates that the electrical conductivity of the plasma channel changes with airflow.

Open-circuit measurements presented in Fig. 3 show a stable high-voltage output with negligible fluctuations, indicating that the power supply operates in a steady-state regime under non-load conditions. The nearly constant voltage waveform confirms the DC nature of the source in the absence of plasma formation.

In contrast, once the arc is initiated, the voltage and current waveforms shown in Fig. 4 exhibit highly oscillatory and nonlinear behavior. The voltage signal displays periodic drops, while the current shows corresponding transient peaks. These measurements were obtained under an air flow velocity of 4 m/s, characteristic of a gliding arc discharge configuration, where the plasma column is continuously stretched and displaced by the gas flow. This airflow contributes to the elongation, instability, and repetitive re-ignition of the arc channel along the electrodes. The observed behavior is attributed to the dynamic nature of the plasma channel, whose effective resistance rapidly changes due to ionization processes, thermal expansion, and the formation of filamentary structures. The imposed airflow further enhances these dynamics by promoting arc motion and repeated breakdown events. In particular, the gas flow increases the rate at which the arc is elongated and quenched, leading to more frequent re-ignition cycles. As a result, the frequency and density of voltage drops and current spikes increase with airflow, since each spike corresponds to a local breakdown or filament formation event. This indicates that the airflow not only controls the spatial evolution of the arc but also governs its temporal characteristics, effectively modulating the repetition rate of discharge events. From an analytical perspective, the spike repetition frequency can be associated with the characteristic time required for arc elongation and re-breakdown, which decreases with increasing flow velocity due to enhanced convective transport and faster removal of charged species and heat from the discharge region. Consequently, higher flow velocities lead to shorter discharge cycles and a higher number of transient events per unit time. The voltage drops correspond to successive breakdown

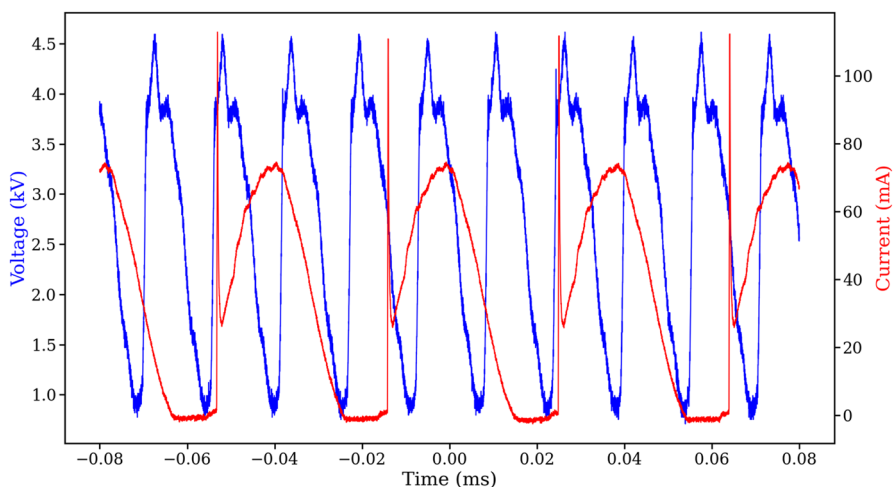


Fig. 4 Temporal evolution of voltage and current in a direct current gliding arc discharge at an airflow velocity of 4 m/s

and re-ignition processes, while the current peaks indicate enhanced conduction through transient plasma filaments. This confirms the presence of a negative differential resistance characteristic, which is typical for atmospheric pressure gliding arc discharges. Although the power supply operates in DC mode, the presence of the gliding arc plasma introduces dynamic instabilities due to the nonlinear and time-varying resistance of the discharge. As a result, the measured voltage and current waveforms in Fig. 4 exhibit oscillatory behavior superimposed on a DC component. This behavior does not indicate AC operation, but rather reflects the intrinsic instability and continuous regeneration of the plasma channel in gliding arc systems under forced airflow conditions.

As airflow increases, the plasma channel transforms into a finer filament structure, and the average conductivity of the plasma column decreases. This is consistent with the observed decrease in both voltage and current values. Electrical measurements also show that the plasma can operate stably at low current levels.

Figure 5 shows the discharge voltage and current values measured as a function of air-flow velocity. The results show that as the airflow increases, the discharge voltage decreases due to the cooling and elongation of the plasma channel, and consequently, the current drawn also decreases.

Gas Temperature and Cathode Temperature Behavior

Gas temperature during plasma formation was measured using a K-type thermocouple. The thermocouple was positioned approximately 1 cm from the electrode region. To minimize electrical interference from the plasma's electric field, the thermocouple was insulated with a silicon tube. The measured gas temperature values showed significant variations depending on the airflow velocity. At low airflow velocities, the gas temperature was measured at approximately 280 °C. However, with increasing airflow velocity, the gas temperature decreased significantly, reaching approximately 32 °C at the maximum airflow velocity. These results indicate the formation of a strong convective cooling mechanism in the plasma channel. Increased airflow rapidly removes energy from the plasma channel, preventing overheating of the gas. This significant decrease in gas temperature also affects the thermal equilibrium of the plasma and contributes to the strengthening of the plasma's non-thermal

Fig. 5 The voltage-current values change depending on the airflow velocity during discharge formation

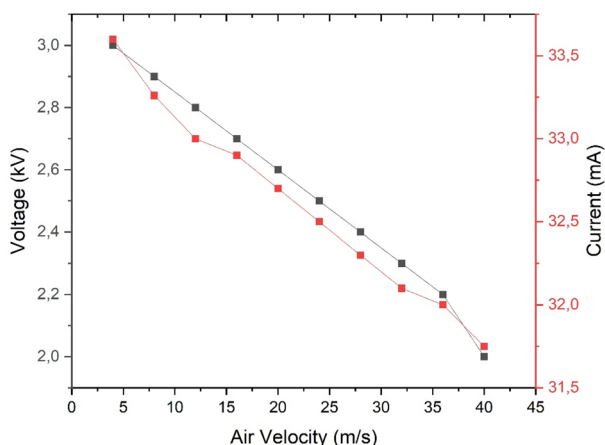


Fig. 6 Changes in gas temperature and cathode temperature during discharge depending on air flow velocity

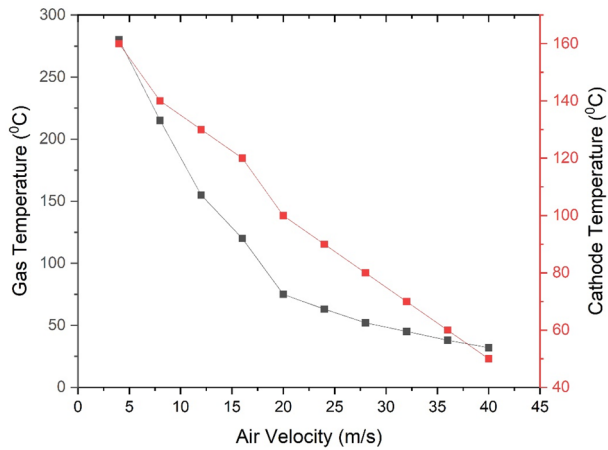


Table 2 Changes in gas temperature and cathode temperature during discharge at different airflow rates

Air Velocity (m/s)	T _{gas} (°C)	T _{gas} (K)	T _{cathode} (°C)
4	280	553	160
8	215	488	140
12	155	428	130
16	120	393	120
20	75	348	100
24	63	336	90
28	52	325	80
32	45	318	70
36	38	311	60
40	32	305	50

character. Cathode temperature was measured non-contact using a mobile thermal camera. Cathode temperature exhibited similar behavior to gas temperature, as electrons were escaping from the cathode and moving towards the anode.

In physics, this phenomenon is described as “cathode fall.” The negative electrode exhibited a higher temperature compared to the positive electrode. This behavior is attributed to the cathode fall region, where a significant portion of the discharge voltage drops. In this region, accelerated positive ions strike the cathode surface, transferring kinetic energy and causing the cathode to heat up further. This event can also be described as ion bombardment. The cathode temperature varied between 50 °C and 160 °C. Therefore, it exhibited a similar behavior to the gas temperature depending on the airflow velocity. In addition, high-speed airflow suppresses the ionization rate and cathode heating while expanding the active excitation volume. Figure 6 shows the variation of gas temperature and cathode temperature during discharge depending on the airflow velocity.

Table 2 shows the variation of gas temperature measured by thermocouple and cathode temperature measured by thermal camera at different airflow velocities during discharge formation. Gas temperature and cathode temperature decreased inversely proportional to airflow velocity.

Figure 7 shows the results of cathode temperature measurements using a thermal camera during discharges created at different airflow velocities. At low flow velocities (4–8 m/s), a large thermal region dominated by high temperatures is observed on the cathode surface, while as the flow velocity increases, the area of this region narrows and the maximum temperature values decrease significantly. At 4 m/s, intense and homogeneous heating is observed at the cathode, while at 20 m/s and above, only localized heating points remain around the arc root, and at 36–40 m/s, the heating almost completely disappears. This indicates that the increased airflow enhances the convective cooling effect, reducing the thermal energy transferred to the cathode and decreasing the intensity of the plasma-electrode interaction. This decrease in cathode temperature is another important indicator of the system's

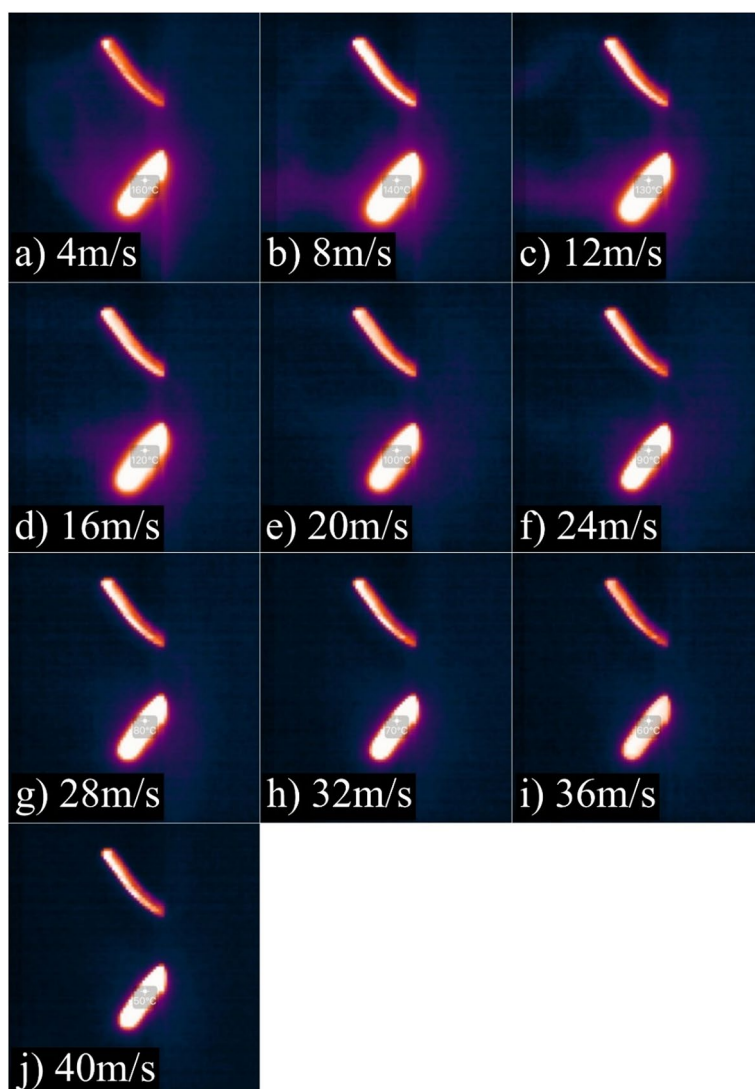


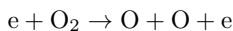
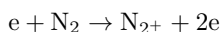
Fig. 7 Variation of cathode temperature at different air flow rates during discharge

transition from a thermal to a non-thermal regime; while thermal effects are dominant at low flow velocities, the plasma acquires a cooler character at high velocities, which can provide advantages in terms of electrode life and energy efficiency. These data reveal that in the GAD system, the flow rate directly controls not only the plasma morphology but also the electrode thermal load.

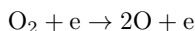
Optical Emission Spectra

The optical properties of the plasma were analyzed using an optical emission spectrometer operating in the 190–1200 nm range. The integration time was set to 400ms when performing optical emission analyses. The resulting spectrum showed the dominance of molecular nitrogen bands, typically observed in atmospheric air plasmas. The strongest emission in the spectrum was observed at approximately 337 nm. This band corresponds to the Second Positive System ($C^3\Pi_u \rightarrow B^3\Pi_g$) transitions of the nitrogen molecule. In addition, First Negative System ($B^2\Sigma_u^+ \rightarrow X^2\Sigma_g^+$) bands belonging to the nitrogen molecule ion were also observed at approximately 391 nm. The intensity of the 337 nm band increased significantly with increasing airflow velocity. This increase can be attributed to the airflow separating the plasma channel into filaments and increasing the total active plasma volume. The 391 nm emission, however, showed lower intensity compared to the 337 nm band. This indicates that the ionization rate in the plasma is limited and excitation processes are more dominant. In the optical emission spectra obtained from the shifted arc plasma, nitrogen molecular bands and nitrogen ion emissions were dominant. The most intense spectral peaks were observed at 337 nm, corresponding to the N_2 second positive system, and at 391 nm, corresponding to the N_2^+ first negative system. The presence of these emission lines indicates strong excitation and ionization processes occurring in the atmospheric air plasma. In addition to nitrogen emissions, weaker spectral lines associated with OH radicals and atomic oxygen were also observed, suggesting the formation of reactive oxygen species in the plasma. Similar emission characteristics have been reported in previous studies investigating atmospheric shifted arc discharges operating in air [36]. Additionally, $NO\gamma$ peaks in the UV region were observed in the spectrum. The reactions occurring during discharge are shown below.

Electron impact reactions



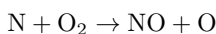
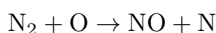
Dissociation reactions



Radical formation



Nitrogen oxide formation



The electron temperature was calculated using the intensity ratio between the 391 nm and 337 nm emission lines. This method is essentially an ionization/excitation ratio [41]. The results show that the electron temperature changes significantly with increasing airflow velocity. At lower airflow velocities, the arc column remained relatively short and concentrated near the electrode base.

Under these conditions, the plasma exhibited lower electron temperatures due to limited arc elongation and reduced plasma volume. As the airflow velocity increased, the arc column elongated along the diverging electrodes, leading to an increase in plasma volume and increased interaction between energetic electrons and gas molecules. This resulted in higher electron temperatures. However, at very high airflow velocities, the electron temperature tended to decrease slightly. This behavior can be attributed to the reduced residence time of the plasma species in the discharge region and the increased convective cooling effect of the airflow. The vibrational temperature of the nitrogen molecules also increased with increasing airflow velocity. This behavior is related to the improved coupling between the plasma and the flowing gas. Higher airflow velocities increase the excitation of nitrogen molecules by creating stronger arc stress and turbulence in the plasma region. As a result, the vibrational temperature of the plasma increases. These findings are consistent with previous experimental studies of atmospheric gliding arc plasmas where vibrational temperatures have been reported to increase with gas flow velocity [36].

Figure 8 shows the optical emission spectrum. Second Positive System bands, belonging to nitrogen molecules, are predominantly observed in the spectrum. The strongest emission is seen around 337 nm, while First Negative System bands belonging to nitrogen ion molecules are detected around 391 nm. Despite using a spectrometer with a range of 190–1200 nm, the 656 nm (H_α), 486 nm (H_β), 777 nm (OI), and 844 nm (OI) lines were not observed in the spectrum. This is because oxygen atoms in the air quickly transform into OH and NO. Even with some humidity in the air (40%), Balmer lines of hydrogen atoms were not observed. This can be attributed to the short lifetimes of the lines that were formed and their rapid formation into OH and NO lines. Furthermore, the absence of copper peaks in the spectrum indicates minimal electrode wear. This suggests that the temperature generated during discharge did not detach the copper atoms from the surface. Considering that the melting point of copper under atmospheric pressure is approximately 1085 °C, this is not the case compared to the maximum cathode temperature of 160 °C. The copper forming the electrodes has not undergone ionization.

When the optical emission spectrum is examined, emissions of various atomic and molecular species characteristic of atmospheric air plasma are observed. The spectrum shows a particular dominance of band systems belonging to the nitrogen molecule. The strongest emission is the N_2 Second Positive System ($\text{C}^3\Pi_u \rightarrow \text{B}^3\Pi_g$) transition observed at approximately 337 nm wavelength. This band is one of the most common emissions in atmospheric air plasmas and is formed by electrons colliding with N_2 molecules, exciting

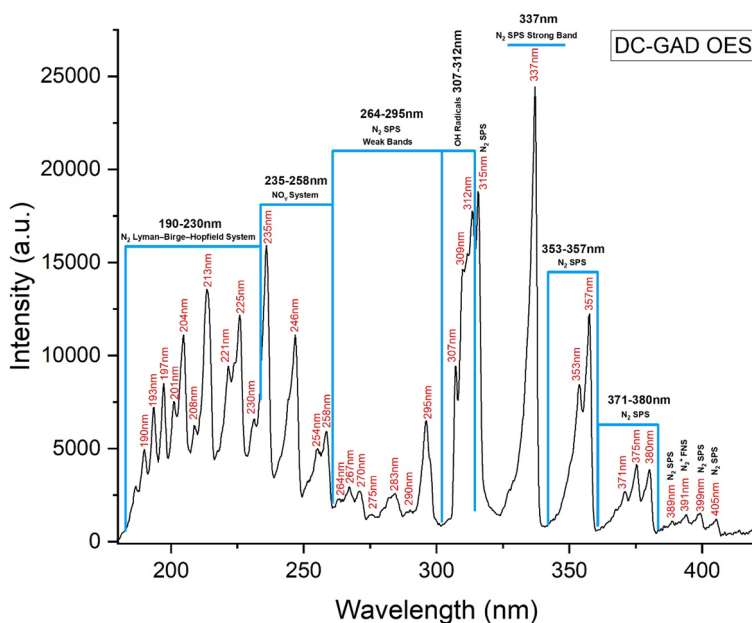
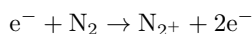


Fig. 8 Optical emission spectrum of atmospheric DC GAD air plasma

them to higher energy levels. This indicates that electron impact excitation is one of the dominant processes in the plasma (Table 3).

The observation of N_2^+ First Negative System ($B^2\Sigma_u^+ \rightarrow X^2\Sigma_g^+$) bands around 391 nm in the spectrum also indicates that ionization processes are taking place in the plasma. These emissions generally result from high-energy electrons ionizing N_2 molecules.

This process can be described by the following reaction:



This result indicates that a certain level of ionization and charge carrier production occurs in the plasma. The NO γ system (235–258 nm) emissions observed in the ultraviolet region of the spectrum are the result of reactions occurring between nitrogen and oxygen atoms in the plasma. In atmospheric air plasmas, electron collisions cause the breakdown of N_2 and O_2 molecules, creating free N and O atoms. These atoms can then combine to form NO molecules.

These processes can be explained by the following reactions:

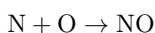
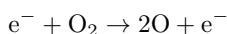
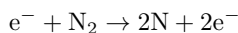
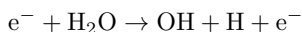


Table 3 Identification of the main emission peaks observed in the DC gliding arc air plasma spectrum together with the corresponding electronic transitions and formation reactions

Wavelength (nm)	Species	Electronic Transition	Spectral System	Main Formation Reaction	References
190–230	N ₂	a ¹ Π _g → X ¹ Σ _g ⁺	N ₂ LBH	e ⁻ + N ₂ → N ₂ * + e ⁻	[6,
204	N ₂	LBH band	N ₂ LBH	e ⁻ + N ₂ → N ₂ (a ¹ Π _g) + e ⁻	34,
213	N ₂	LBH band	N ₂ LBH	electron impact excitation	37,
221–225	N ₂	LBH band	N ₂ LBH	e ⁻ + N ₂ → N ₂ * + e ⁻	40,
235	NO	A ² Σ ⁺ → X ² Π	NO system	N + O → NO*	42,
246	NO	γ band	NO system	N ₂ + O → NO + N	55]
254	NO	γ band	NO system	N + O ₂ → NO + O	
258	NO	γ band	NO system	excited NO formation	
264–275	N ₂	C ³ Π _u → B ³ Π _g	N ₂ SPS	e ⁻ + N ₂ → N ₂ (C ³ Π _u) + e ⁻	
283	N ₂	SPS band	N ₂ SPS	electron excitation	
295	N ₂	SPS band	N ₂ SPS	N ₂ (C) → N ₂ (B) + hν	
307	OH	A ² Σ ⁺ → X ² Π	OH(A–X)	H ₂ O + e ⁻ → OH* + H + e ⁻	
309	OH	A ² Σ ⁺ → X ² Π	OH(A–X)	H ₂ O dissociation	
312	OH	A ² Σ ⁺ → X ² Π	OH band	OH* → OH + hν	
315	N ₂	C ³ Π _u → B ³ Π _g	N ₂ SPS	e ⁻ + N ₂ → N ₂ (C)	
337	N ₂	C ³ Π _u → B ³ Π _g	N ₂ SPS	N ₂ (C ³ Π _u) → N ₂ (B ³ Π _g) + hν	
353	N ₂	SPS band	N ₂ SPS	vibrational relaxation	
357	N ₂	C ³ Π _u → B ³ Π _g	N ₂ SPS	radiative decay	
371	N ₂	SPS band	N ₂ SPS	excited nitrogen relaxation	
375	N ₂	SPS band	N ₂ SPS	N ₂ (C) → N ₂ (B)	
380	N ₂	SPS band	N ₂ SPS	vibrational transition	
391	N ₂ ⁺	B ² Σ _u ⁺ → X ² Σ _g ⁺	N ₂ ⁺ FNS	e ⁻ + N ₂ → N ₂ ⁺ + 2e ⁻	
399	N ₂	SPS band	N ₂ SPS	recombination processes	
405	N ₂	SPS band	N ₂ SPS	ion excitation	

Radiative transitions observed during the transition of NO molecules from the excited state to the ground energy level are seen as NOγ bands in the spectrum. This indicates that active nitrogen-oxygen chemistry is taking place in the plasma.

The bands observed in the spectrum around 307–312 nm have been identified as belonging to OH (A²Σ⁺ → X²Π) transitions. OH radicals are generally formed as a result of the dissociation of water vapor present in the atmosphere within the plasma. These reactions can be described as follows:



OH radicals are highly reactive species in atmospheric plasmas and play a significant role, particularly in oxidation reactions. Therefore, the observation of OH emissions in the spectrum indicates that water vapor-induced radical chemistry is also active in the plasma.

In addition, N₂ Lyman–Birge–Hopfield (LBH) bands observed in the 190–230 nm range indicate the presence of high-energy electrons in the plasma. These bands are generally molecular transitions originating from high energy levels and demonstrate that strong electron collision processes are occurring in the plasma.

When all spectral characteristics are evaluated together, it is understood that the DC gliding arc air plasma created in this study involves the following basic chemical processes:

1. Excitation of N_2 and O_2 molecules by electron collisions.
2. Molecular breakdown processes by electron collisions.
3. N_2 ionization and formation of N_2^+ species.
4. Formation of NO as a result of the reaction of N and O atoms.
5. Formation of OH radicals as a result of the decomposition of water vapor.

These reactions are indicative of the typical N_2 – O_2 – H_2O plasma chemistry seen in atmospheric plasmas. The obtained spectral data show that the gliding arc plasma supports not only ionization processes but also a rich reactive radical chemistry.

This provides a significant advantage in terms of the potential use of gliding arc plasmas in fields such as environmental applications, gas conversion technologies, and plasma-assisted chemical reactions.

Figure 9 shows a comparison of OES results generated at different airflow velocities. A careful examination of this figure reveals an increase in the intensity of the peaks with increasing airflow velocity. The data in Table 4 show that the spectral line intensities in the DC gliding arc plasma change significantly depending on the airflow velocity. Examining the general trend, it is observed that emission intensities are relatively low at low airflow velocities (4–12 m/s), and a significant increase occurs for many spectral lines, especially in the 16–32 m/s range, as the airflow velocity increases. This increase can be attributed to the strengthening of electron-molecule collisions and excitation processes as the amount of gas transported to the plasma region increases. In particular, the 315 nm, 337 nm, and 357 nm bands belonging to the N_2 Second Positive System, and the OH radical band (~307–309 nm) reach maximum values in this range, indicating increased effective excitation and reactive species production in the plasma. However, it is observed that the emission intensity decreases again in many lines as the airflow velocity reaches higher values (36–40 m/s). This can be explained by the high gas flow cooling the plasma column, the rapid removal of reactive species and excited states from the plasma region by convective transport, and the transition of the discharge to a more filamentary and less dense regime. Therefore, the results show that airflow velocity is an important parameter directly controlling the exci-

Fig. 9 Comparison of OES graphs of discharge generated at different airflow velocities

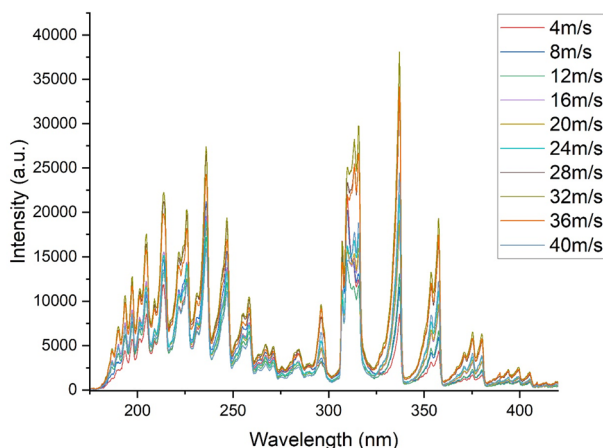


Table 4 Comparison of the intensity (a.u.) values of the lines in the spectrum depending on different airflow velocities in the OES graph

Wavelength (nm)	4 m/s	8 m/s	12 m/s	16 m/s	20 m/s	24 m/s	28 m/s	32 m/s	36 m/s	40 m/s
190 (N ₂)	2311	3097	4121	5149	4812	4696	6737	7065	6672	4777
193 (N ₂)	4386	6056	6254	7577	7080	6955	9866	10,631	9783	7222
197 (N ₂)	5710	7560	7217	8975	8473	8498	12,101	12,612	11,567	8288
201 (N ₂)	5771	7411	6865	8206	7867	7642	10,903	11,319	10,656	7530
204 (N ₂)	8576	10,723	10,033	12,194	11,705	11,590	16,397	17,538	15,780	11,109
208 (N ₂)	5540	6722	5825	6926	6625	6860	9637	10,275	9135	6213
213 (N ₂)	11,867	14,612	13,217	15,556	15,015	15,174	21,208	22,233	19,830	13,491
221 (N ₂)	9454	11,289	9206	11,087	10,576	10,821	14,906	15,554	14,045	9436
225 (N ₂)	11,916	14,355	12,491	14,405	13,738	14,221	19,687	20,293	18,180	12,165
230 (N ₂)	7070	8161	6590	7730	7427	7582	10,521	10,817	10,069	6402
235 (NO)	18,184	21,204	17,151	19,545	18,965	19,010	26,550	27,399	24,270	15,920
246 (NO)	13,545	15,646	12,107	13,804	13,357	13,341	18,610	19,384	16,956	11,115
254 (NO)	6294	7143	5352	6204	6006	5935	8414	8541	7659	4954
258 (NO)	7586	8622	6623	7526	7209	7351	10,195	10,481	9306	5903
264 (N ₂)	3021	3739	2440	2812	2731	2842	3830	4016	3526	2211
267 (N ₂)	3918	4228	3254	3682	3551	3669	5065	5183	4560	2945
270 (N ₂)	3626	4052	3120	3483	3289	3345	4748	4840	4369	2706
275 (N ₂)	2109	2171	1705	1866	1930	1934	2520	2671	2416	1466
283 (N ₂)	4129	4004	2753	3250	3103	3130	4422	4531	3937	2534
290 (N ₂)	2679	2535	1875	2130	1991	2092	2815	3025	2620	1708
295 (N ₂)	3049	3328	2978	3995	4195	4767	7301	8177	7943	5634
307 (OH)	16,804	15,182	10,200	11,293	11,143	11,096	15,935	16,622	14,383	9416
309 (OH)	21,994	20,231	14,281	16,132	15,812	16,273	23,267	25,064	21,701	14,628
312 (OH)	14,647	14,378	11,418	13,890	14,128	15,201	22,633	24,906	22,144	15,373
315 (N ₂)	12,408	13,097	11,734	15,239	15,971	17,582	26,642	29,730	26,608	18,814
337 (N ₂)	8559	11,590	13,098	18,334	19,057	21,954	33,403	38,106	34,148	24,429
353 (N ₂)	3157	4192	4704	6365	6779	7662	11,877	13,278	11,878	8393
357 (N ₂)	4398	5949	6672	9235	9716	11,147	16,970	19,313	17,451	12,230
371 (N ₂)	1190	1452	1445	2092	2116	2463	3668	4206	3915	2651
375 (N ₂)	1709	2030	2328	3097	3337	3721	5704	6522	5830	4146
380 (N ₂)	1673	2105	2280	3080	3228	3704	5613	6330	5786	3885
389 (N ₂)	1303	1082	779	936	1056	1151	1535	1690	1965	874
391 (N ₂ ⁺)	724	771	684	861	871	987	1435	1588	1527	1062
399 (N ₂)	922	1016	968	1316	1337	1505	2188	2555	2337	1535
405 (N ₂)	911	841	870	1026	1025	1264	1750	2048	1901	1227

tation kinetics and plasma chemistry of DC gliding arc plasma, and that plasma activity reaches its maximum at moderate flow rates. It should be emphasized that the observed increase in emission intensity with airflow velocity cannot be solely attributed to an increase in plasma volume. Optical emission spectroscopy (OES) measurements represent line-of-sight integrated signals and are influenced by multiple factors, including plasma morphology, local excitation conditions, and spatial distribution of emitting species. Although the arc length decreases with increasing airflow velocity (Fig. 16), the transition from a continuous arc to a filamentary discharge leads to the formation of multiple micro-discharge channels. These filaments can enhance local electron impact excitation processes, resulting in

Table 5 Average emission intensities of selected spectral lines in different plasma regimes ($\pm$ SD)

Spectral Lines (Species)	Arc Regime (4–8–12 m/s) $n=3$	Transition Regime (16–20 m/s) $n=2$	Filamentary Regime (24–28–32–36– 40 m/s) $n=5$
337 nm (N_2)	11,082 $\pm$ 2274	18,696 $\pm$ 511	30,408 $\pm$ 6745
391 nm (N_2^+)	726 $\pm$ 44	866 $\pm$ 7	1252 $\pm$ 310
235 nm (NO)	18,846 $\pm$ 2036	19,255 $\pm$ 410	22,230 $\pm$ 5483
309 nm (OH)	18,836 $\pm$ 4025	15,972 $\pm$ 370	20,177 $\pm$ 4680
357 nm (N_2)	5673 $\pm$ 1145	9476 $\pm$ 340	15,422 $\pm$ 3828
315 nm (N_2)	12,413 $\pm$ 657	15,605 $\pm$ 518	23,845 $\pm$ 4971

Table 6 The effect size of the change in emission intensities during the transition from arc regime to filamentary regime (Cohen's d)

Spectral Lines (Species)	Arc $\rightarrow$ Filamentary Variation (%)	Cohen's d	Effect Size
337 nm (N_2)	%174 increase	3.12	Very large
391 nm (N_2^+)	%72 increase	1.95	Large
235 nm (NO)	%18 increase	0.72	Moderate
309 nm (OH)	%7 increase	0.29	Small
357 nm (N_2)	%172 increase	2.84	Very large
315 nm (N_2)	%92 increase	2.58	Very large

increased overall emission intensity despite a reduction in the macroscopic plasma length. In addition, variations in electron density and electron energy distribution with airflow contribute to changes in excitation efficiency.

It should also be noted that spectroscopic measurements are based on relative intensity values expressed in arbitrary units. Therefore, the OES analysis presented in this study has been interpreted as semi-quantitative rather than qualitative, and the observed trends have been evaluated relatively, not absolutely.

Optical emission spectra obtained in different plasma regimes (arc, transition, filamentary) demonstrate that regime change has a significant effect on spectral line intensities. To evaluate the effect of plasma regimes on optical properties, the peaks N_2^+ 337 nm, N_2^+ 391 nm, NO 235 nm, and OH 309 nm, which are dominant in atmospheric air plasmas and represent excitation processes, ionization degree, and reactive nitrogen and oxygen species respectively, as well as the peaks N_2 357 nm and N_2 315 nm used in vibrational temperature calculations, were selected for detailed analysis. The descriptive statistical results presented in Table 5 reveal that the emission intensities of all selected spectral lines tend to increase from the arc regime to the filamentary regime.

Table 6 shows the effect size (Cohen's d) values of the change in emission intensities during the transition from the arc regime to the filamentary regime.

The most significant increase was observed in N_2 (337 nm) emission. While the average intensity in the arc regime was 11082 $\pm$ 2274 a.u., it increased by approximately 174% to 30408 $\pm$ 6745 a.u. in the filamentary regime. The effect size of this increase was calculated as Cohen's $d=3.12$, which is classified as a 'very large effect' in the literature. This finding shows that electron-molecule collisions and excitation processes are significantly increased in the filamentary regime.

N_2^+ (391 nm) ion emission also showed a remarkable increase during the regime transition. The average intensity, which was 726 $\pm$ 44 a.u. in the arc regime, increased by 72% to

1252 ± 310 a.u. in the filamentary regime. This increase, calculated as Cohen's $d=1.95$, is at the 'large effect' level, confirming that ionization processes and the presence of high-energy electrons are increased in the filamentary regime.

NO (235 nm) emission was measured as 18846 ± 2036 a.u. in the arc regime and 22230 ± 5483 a.u. in the filamentary regime. The effect size of this 18% increase is at the 'medium-large' level with Cohen's $d=0.72$. This result indicates that nitrogen-oxidation reactions ($N+O \rightarrow NO$) become more active in the filamentary regime.

OH (309 nm) radical emission, on the other hand, showed a more limited increase compared to other species in the regime transition (arc: 18836 ± 4025 a.u., filamentary: 20177 ± 4680 a.u.; 7% increase). This increase, remaining at the 'small effect' level with Cohen's $d=0.29$, suggests that water vapor-induced reactions are relatively less affected by the regime change.

The N_2 (357 nm) and N_2 (315 nm) emissions used in vibrational temperature calculations also increased by 172% (Cohen's $d=2.84$) and 92% (Cohen's $d=2.58$), respectively, both increases being at the 'very large effect' level.

In the transition regime, all spectral lines exhibited an intermediate value between arc and filamentary regimes. This indicates that the transition regime has a mixed character, combining features of both thermal arc and non-thermal filamentary plasma.

Emission Intensity Ratio and Electron Energy Characteristics

To evaluate the ionization and excitation processes in the plasma, the I_{391}/I_{337} emission density ratio was calculated. This ratio provides important information about the degree of ionization and electron energy distribution in the plasma. The results showed that this ratio does not remain constant depending on the airflow velocity. At low airflow velocities, the ratio was measured at approximately 0.03, while at medium flow rates it decreased significantly and showed a fluctuating behavior in the range of approximately 0.002–0.005. At higher airflow velocities, a partial increase was observed again. This behavior indicates that the ionization and excitation processes in the plasma are significantly affected by the airflow. In particular, at low flow rates, the plasma behaves more thermally, while at increasing flow rates, the plasma separates into filaments and the electron energy distribution changes. Therefore, airflow emerges as a significant control parameter on the electron energy distribution function (EEDF) in the plasma.

Figure 10 shows a comparison of the intensity values of the 391 nm and 337 nm spectral lines as a function of different airflow velocities.

Electron Temperature Variation

The average electron temperature in the plasma was estimated using optical emission data. The calculated electron temperature values ranged from approximately 1.4 eV to 2.8 eV. The highest electron temperature was observed at low airflow velocities. It was determined that the electron temperature generally tended to decrease with increasing airflow. The main reason for this is that the increased gas flow reduces the energy density in the plasma channel and the plasma transforms into a filamentary structure. A significant correlation was observed between the change in electron temperature and the change in the I_{391}/I_{337} ratio.

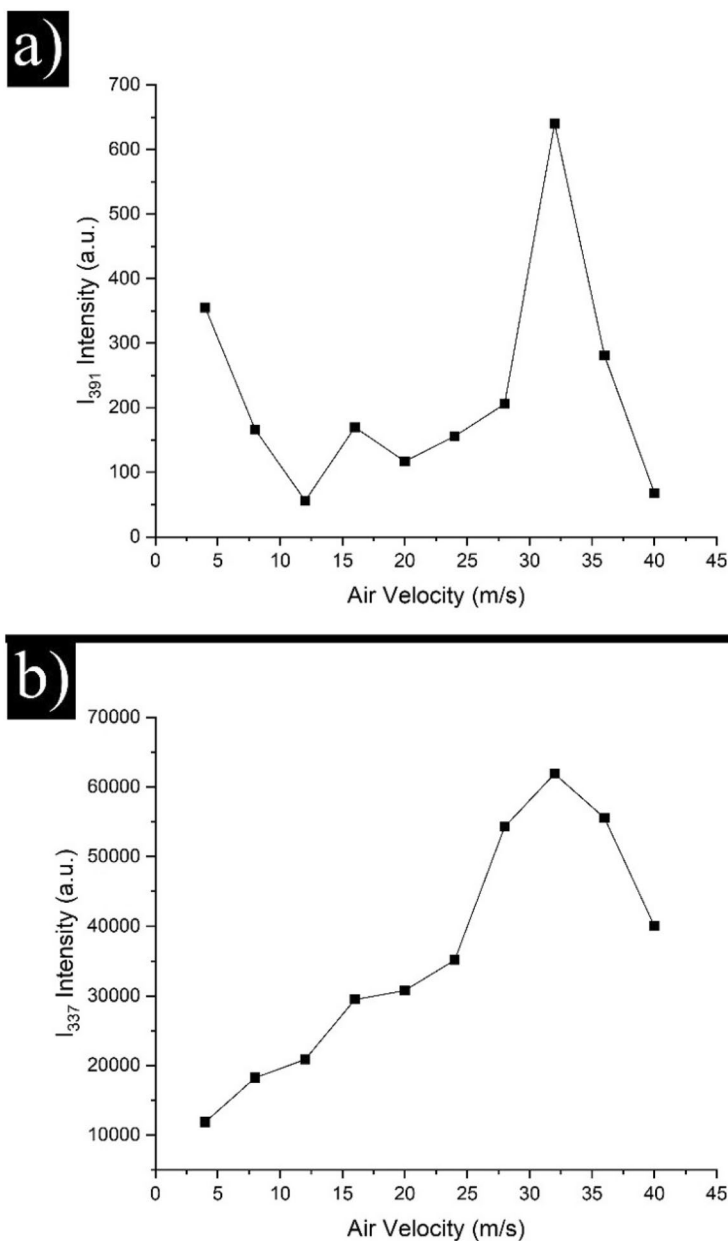


Fig. 10 Comparison of the intensity values of (a) 391nm and (b) 337nm spectral lines using OES at different airflow velocities during discharge

This demonstrates that the electron energy distribution is directly related to the airflow. The approximate electron temperature equivalents of the obtained ratios are given from the literature. The conversion between the emission density ratio and electron temperature is based on calibration curves previously reported from Boltzmann equation simulations using

electron collision cross-section data for nitrogen. This diagnostic method is widely used in atmospheric pressure air plasmas because the 391 nm emission originates from electron collision ionization, while the 337 nm band originates from electron collision excitation of molecular nitrogen. Since ionization requires higher electron energies, the I_{391}/I_{337} density ratio becomes highly dependent on the electron energy distribution. The relationship between this ratio and electron temperature is derived from previously reported collision-radiation models and Boltzmann simulations found in the literature [34, 41].

Table 7 shows a comparison of the I_{391}/I_{337} ratio and electron temperatures in discharges obtained at different airflow velocities. It should be noted that the electron temperature values derived from the I_{391}/I_{337} emission ratio should be interpreted with caution. Under atmospheric pressure conditions, where collisional processes are dominant, the obtained values are more accurately representative of an excitation temperature (T_{exc}) rather than the true electron temperature (T_e) associated with the electron energy distribution function (EEDF). Furthermore, the calibration curves used in this study were originally derived from low-pressure plasma conditions, which differ significantly from the highly collisional environment of atmospheric gliding arc discharges. Therefore, the reported values should be considered as approximate indicators of electron energy rather than absolute thermodynamic electron temperatures. This approach implicitly assumes a quasi-equilibrium between selected excited states and a stable electron energy distribution, conditions that may not be fully satisfied in highly transient and non-equilibrium atmospheric gliding arc discharges. Nevertheless, the method remains useful for capturing relative trends in electron energy variations under different operating conditions.

Figure 11 shows a graphical comparison of the I_{391}/I_{337} ratio and electron temperatures in discharges obtained at different airflow velocities.

Vibrational Temperature Analysis

To determine the molecular energy distribution of the plasma, the vibrational temperature was calculated using the Boltzmann plot method [72]. This calculation utilized the 337 nm, 357 nm, and 380 nm transitions of the Second Positive System bands of the nitrogen molecule. The calculated vibrational temperature values were found to be approximately in the range of 5×10^3 K to 1.6×10^4 K.

These values are quite high compared to the gas temperature, indicating that the plasma has a strong thermal instability character. This suggests that atmospheric gliding arc discharges can exhibit both thermal and non-thermal plasma properties simultaneously.

Table 7 Numerical comparison of I_{391}/I_{337} ratio and electron temperatures in discharges obtained at different airflow velocities

Air velocity (m/s)	I_{337} (Area)	I_{391} (Area)	I_{391}/I_{337}	T_e (eV)
4	11,904	355	0.029	2,8
8	18,264	166	0.009	2,2
12	20,888	56	0.002	1,6
16	29,501	170	0.005	1,9
20	30,780	117	0.003	1,7
24	35,159	156	0.004	1,8
28	54,291	206	0.003	1,7
32	61,901	640	0.010	2,2
36	55,558	281	0.005	1,9
40	40,062	68	0.001	1,4

Fig. 11 Graphical comparison of I_{391}/I_{337} ratio and electron temperatures in discharges obtained at different airflow velocities

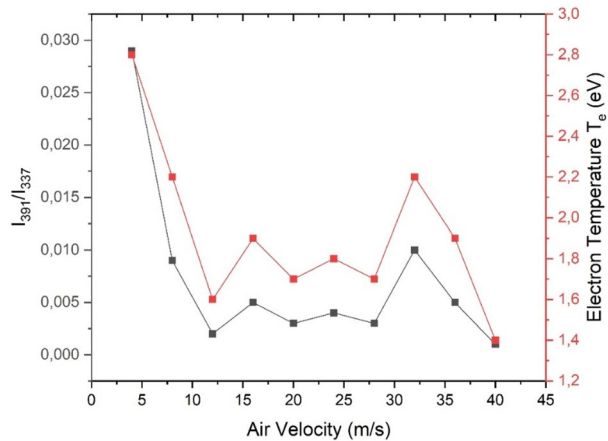


Table 8 Parameters used for calculating vibrational temperature values [72]

λ (nm)	Area	A	E_v (eV)
337	A1	1	0
357	A2	0.56	0.29
380	A3	0.28	0.58

Table 8 shows the parameter table used to calculate the vibrational temperature values of discharges obtained at different airflow velocities. In the table, the term expressed as Area represents the areas under the specified spectral lines. The term A represents the Einstein coefficient used for the specified spectral lines. The term E_v represents the vibrational energy level used for the specified spectral lines. To calculate the vibrational temperature value, the equation $\ln(I\lambda/A)$ was first calculated separately for each of the specified spectral lines. Then, a scattering plot was drawn where the results from this equation formed the y-axis and the E_v values formed the x-axis. The slope values were calculated for the gas flow by performing linear fitting on the graph.

The obtained slope values were converted into vibrational temperature values using the equation $T_v = (-1/km)$. In the equation, the term k represents the Boltzmann constant and the term m represents the slope. Table 9 shows a comparison of the vibrational temperature values of the discharges obtained at different airflow velocities. The fitting parameters, including slope, intercept, and R^2 values, are summarized in Table 9. The variability observed in vibrational temperatures is consistent with the deviations from ideal linearity discussed above, particularly at airflow conditions with lower R^2 values.

Figure 12 shows a comparison of the linearly fitted results of the discharge graphs $\ln(I\lambda/A)$ vs. E_v obtained at different airflow velocities. The quality of the Boltzmann plot fitting was evaluated using the coefficient of determination (R^2), with values ranging between 0.85 and 0.99, as shown in Fig. 12. In most cases, a strong linear correlation between $\ln(I\lambda/A)$ and vibrational energy levels was observed, confirming the applicability of the Boltzmann method under the present experimental conditions. However, slight deviations from ideal linearity were detected at certain airflow velocities, which may contribute to the fluctuations observed in the calculated vibrational temperatures. These variations can be attributed to several factors, including the limited number of spectral lines used in the analysis, measure-

Table 9 Comparison of vibrational temperature values of discharges obtained at different airflow velocities

Air Velocity (m/s)	Slope	R^2	Intercept	T_v (K)
4	-0.7283	0.9973	15.3006	15,934 K
8	-2.28517	0.85731	15.78887	5078 K
12	-0.81982	0.97531	15.78885	14,154 K
16	-1.78663	0.89586	16.21425	6495 K
20	-1.96441	0.87782	16.27741	5907 K
24	-0.81338	0.98867	16.3023	14,265 K
28	-0.8856	0.9699	16.74832	13,103 K
32	-0.86318	0.97215	16.87784	13,443 K
36	-1.66347	0.89391	16.84122	6974 K
40	-0.95988	0.99177	16.43291	12,088 K

ment uncertainties associated with emission intensities, and the inherently unstable and non-uniform structure of the gliding arc plasma, particularly in the transition and filamentary regimes. In addition, the dynamic nature of the discharge and the spatial inhomogeneity of the plasma may lead to variations in local excitation conditions, affecting the reliability of the linear fitting. Therefore, the reported vibrational temperatures should be interpreted as approximate values that reflect general trends in the plasma behavior rather than exact thermodynamic quantities.

Figure 13 shows a graphical comparison of the vibrational temperature values of discharges obtained at different airflow velocities. The results presented in the figure reveal that the vibrational temperature (T_v) in the DC gliding arc plasma exhibits significant fluctuations depending on the airflow velocity. At a low airflow velocity (4 m/s), the vibrational temperature reaches a fairly high value of approximately 1.6×10^4 K, while a significant decrease is observed as the airflow velocity increases to 8 m/s, falling to approximately 5×10^3 K. This can be explained by the fact that the increased gas flow enhances energy transfer and convective cooling effects in the plasma region, partially suppressing the vibrationally excited states. However, with a renewed increase in airflow velocity in the range of 12–32 m/s, the vibrational temperature rises again and reaches approximately 1.3 – 1.4×10^4 K. This range indicates the operating regime where the plasma column is more stable and electron-molecule collisions effectively sustain vibrational excitation. At higher airflow velocities (36–40 m/s), however, the vibrational temperature shows a fluctuating trend again. This may be related to effects such as elongation of the plasma column, increased filamentary structure, and more heterogeneous distribution of energy density in space at high gas velocities. Overall, the results show that airflow velocity is a parameter that strongly influences the vibrational energy distribution in DC gliding arc plasma and that the plasma system maintains its non-thermal character ($T_v \gg T_g$).

Table 10 compares the ranges of electron, gas, and vibrational temperature values of discharges obtained at different airflow velocity values. Looking at the table, the situation is $T_e > T_v > T_g$. This indicates that the obtained discharge is generally a non-equilibrium plasma.

Plasma Regime Transition

When electrical measurements, optical emission analyses, and temperature measurements are evaluated together, it is seen that airflow velocity is a significant control parameter on the plasma regime. At low airflow velocities, the plasma exhibits a more stable arc regime. With increasing airflow, the plasma channel elongates and breaks down, transforming

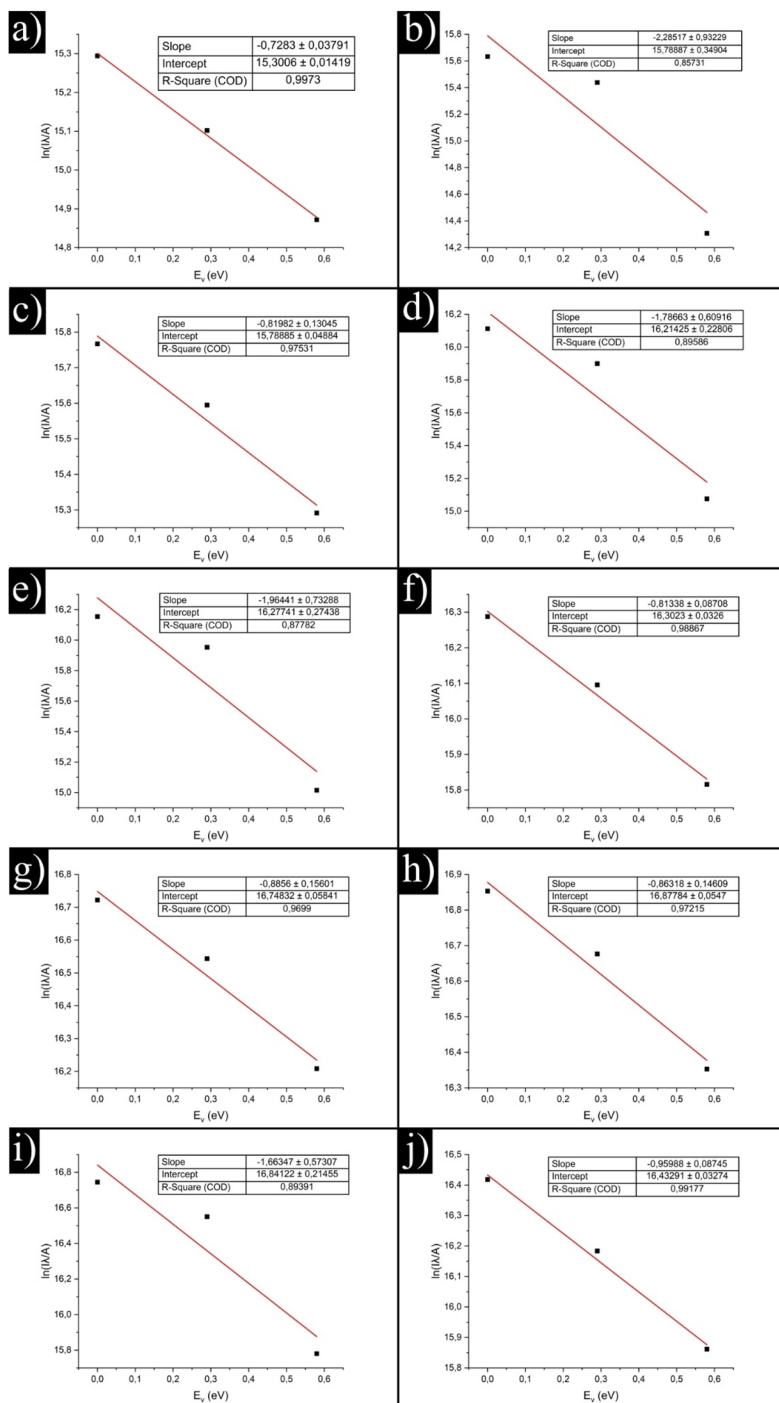


Fig. 12 Comparison of linearly fitted discharge $\ln(I/A)$ vs. E_v graphs obtained at different airflow velocities: (a) 4 m/s, (b) 8 m/s, (c) 12 m/s, (d) 16 m/s, (e) 20 m/s, (f) 24 m/s, (g) 28 m/s, (h) 32 m/s, (i) 36 m/s, (j) 40 m/s

Fig. 13 Graphical comparison of vibrational temperature values of discharges obtained at different airflow velocities

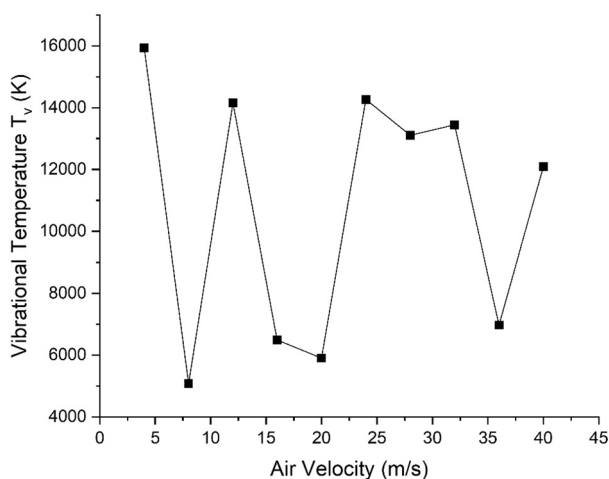


Table 10 Comparison of temperature values, which are fundamental parameters of plasma

Parameter	Value
Electron Temperature (T_e)	16,240–32,480 K
Vibrational Temperature (T_v)	5078–15,934 K
Gas Temperature (T_g)	305–553 K

into a filamentary discharge structure. This behavior is clearly shown with the generated plasma regime map. The map shows the arc regime at low flow rates, the transition region at medium flow rates, and the filamentary plasma regime at high flow rates.

Figure 14 shows the voltage-dependent regime map of discharges obtained at different airflow velocities. Looking at the graph, voltage drops occur at the output of the power supply with increasing airflow velocity. In the thermal region, defined as the Arc Regime in the 4–12 m/s range, the plasma is observed in a stable arc form with high temperature and ionization level, while in the transition regime between 12 and 20 m/s, both thermal and non-thermal characteristics are seen together, with irregularities and cooling effects becoming more pronounced in the plasma filaments. In the 20–40 m/s range, the system completely transitions to a non-thermal filamentary regime; in this region, the plasma is characterized by thin filaments, low temperature, and high electron energy.

Figure 15 shows a real-time view of discharges obtained at different airflow velocities. Analysis of the figure reveals that the discharge pattern changes depending on the airflow velocity. This clearly illustrates the regime transition. A transition pattern from arc initiation to filamentary discharge is observed. At low velocities (4–8 m/s), the plasma channel exhibits a wider, more stable, and denser structure, and the arc remains between the electrodes for a longer period, creating a homogeneous thermal distribution. As the flow rate increases (12–28 m/s), the plasma channel thins and a vibrating structure becomes more pronounced. Energy density increases due to arc cooling via convection, while discharge stability decreases. At velocities of 32 m/s and above, the plasma filaments become quite thin, irregular, and short-lived; the homogeneity of the discharge is lost due to intense turbulence and rapid cooling, which directly affects the distribution of active species and reaction kinetics in terms of plasma chemistry. In addition to convective cooling and plasma

Fig. 14 Creating a voltage-dependent discharge regime map of discharges obtained at different airflow velocities

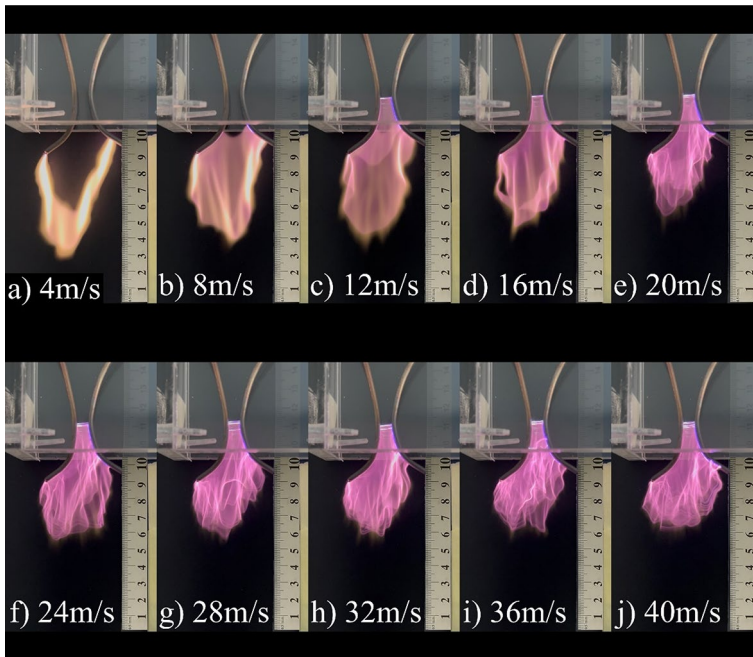
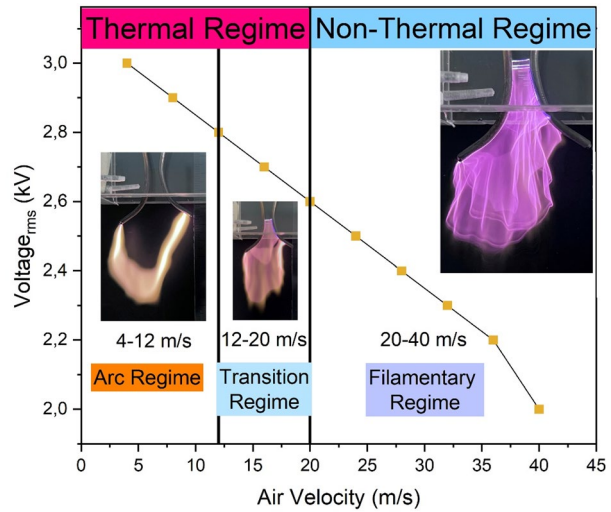


Fig. 15 Real-time view of discharges obtained at different airflow velocities

elongation, the transition from arc to filamentary regime can also be interpreted in terms of changes in the reduced electric field (E/N) and the residence time of plasma species. As the airflow velocity increases, the gas temperature decreases, leading to an increase in gas number density. This results in a reduction of the reduced electric field (E/N), which directly affects the electron energy distribution and reduces ionization efficiency.

Furthermore, higher airflow velocities significantly decrease the residence time of charged and excited species within the discharge region. This limits the duration of energy transfer processes and promotes the formation of short-lived, spatially localized filamentary structures. These combined effects contribute to the observed transition from a stable arc regime to a non-thermal filamentary regime.

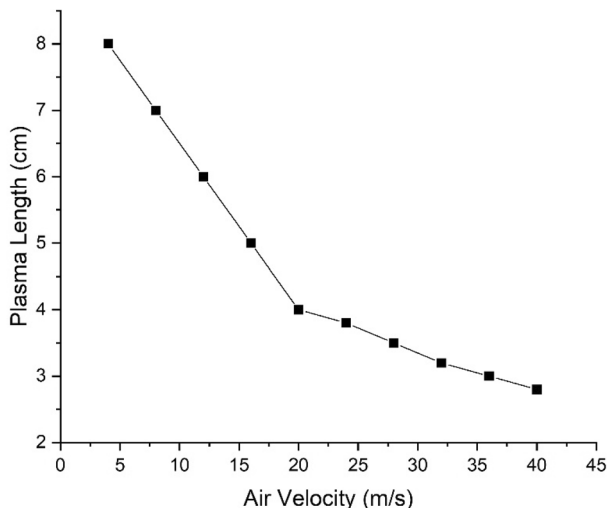
Figure 16 shows the variation in discharge length obtained at different airflow velocities. The results in the figure show that the plasma arc length decreases significantly with increasing airflow velocity. While the arc length is measured as approximately 8 cm at a low airflow velocity (4 m/s), it gradually decreases to approximately 3 cm at 40 m/s with increasing airflow velocity. This can be explained by the strong convective cooling and aerodynamic shear forces that the plasma channel is subjected to at high airflow velocities. The increased gas flow makes it difficult for the plasma column to elongate stably, causing the discharge to occur in a shorter but more dynamic plasma region. In addition, the rapid removal of ionized species and excited particles from the plasma region at high flow rates limits the sustainable length of the plasma column. Therefore, the results show that airflow velocity is a critical parameter that directly affects the geometry and plasma stability of the DC gliding arc discharge. However, when the airflow velocity is excessively high, the arc can become unstable and extinguish prematurely.

Power Consumption and Efficiency

Figure 17 shows the variations in plasma power, fan power, and total power consumption in the GAD system depending on the airflow velocity. As the flow rate increases, the plasma power decreases steadily from 102 W to 68 W, while the fan power increases from 0 to 8 W. Total power consumption also decreases from 105 W to 85 W, indicating that the system consumes less total energy at higher flow rates.

This can be explained by the fact that the increased airflow cools the plasma channel, reducing the discharge resistance and allowing the arc to be maintained at a lower power. In terms of system efficiency, the ratio of plasma power to total power ($\eta = P_{\text{plasma}} / P_{\text{total}}$) can be used as an efficiency indicator.

Fig. 16 Variation in discharge length obtained at different airflow velocities



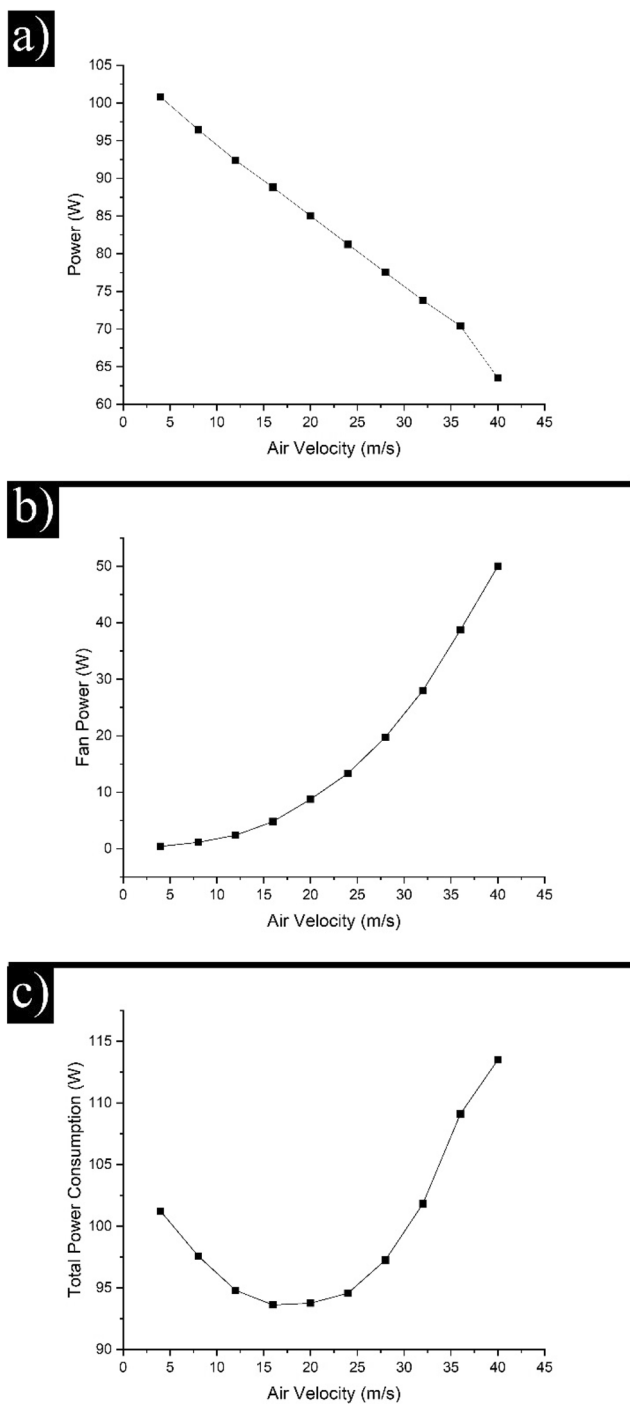


Fig. 17 Graphical comparison of the power consumption of (a) the power supply, (b) the snail fan, and (c) the system as a whole during discharges obtained at different airflow velocities

Table 11 shows the power consumption and efficiency values of discharges generated at different airflow rates. These ratios indicate that at low flow rates, the share of plasma power in the total power is quite high, but as the flow rate increases, the efficiency gradually decreases due to increased fan consumption and decreased plasma power. However, the decrease in absolute plasma power consumption suggests that the system can operate more energy-efficiently at high flow rates, but the optimum operating range should be determined depending on process objectives such as plasma density and active species production. These data clearly reveal a trade-off between energy efficiency and plasma stability in the GAD system.

E/N Reduction

The ratio (E/N) was used to calculate the reduced electric fields of discharges obtained at different airflow velocities. In this ratio, E represents the electric field, while N represents the numerical density of the gas. The gas number density was calculated using the ideal gas relation $N = P/kT$, where the gas temperature measured by thermocouple was converted to Kelvin. Airflow directly controls the reduced electric field (E/N) in gliding arc discharge.

In this equation, P represents atmospheric pressure in Pascals, k represents the Boltzmann constant, and T represents the temperature of the gas in Kelvin at the relevant airflow velocity. After determining the E/N values for each airflow velocity, the Td values were calculated by performing the Townsend transformation ($1 \text{ Td} = 10^{-21} \text{ V} \cdot \text{m}^2$) [74]. Table 12 shows the numerical variation of the E/N (Td) value depending on different airflow velocity ratios.

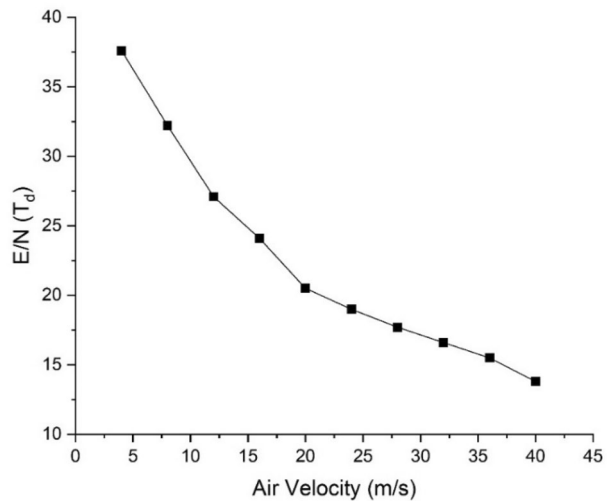
Figure 18 shows that the reduced electric field strength (E/N) decreases regularly with increasing air flow velocity in the GAD system. The E/N value, which is 37.6 Td at 4 m/s, decreases to 13.8 Td at 40 m/s. The main reason for this decrease is that as the flow velocity increases, the gas temperature (T_{gas}) decreases from 553 K to 305 K, and consequently, the gas density (N) increases from $1.33 \times 10^{25} \text{ m}^{-3}$ to $2.41 \times 10^{25} \text{ m}^{-3}$. At the same time, the electric field strength (E) also decreases from $5.00 \times 10^5 \text{ V/m}$ to $3.33 \times 10^5 \text{ V/m}$. The E/N ratio is one of the most important parameters determining the average energy gained by electrons in the plasma and therefore the efficiency of fundamental processes such as ionization, excitation, and dissociation. High E/N values indicate that electrons can reach higher energies, more effectively exciting or breaking down molecules; while low E/N values signal that the plasma has transitioned to a non-thermal regime and that electron energies have decreased relatively. In this context, the decrease in E/N with increasing airflow is one of the key

Table 11 Numerical representation of the power consumption and efficiency values of discharges created at different airflow rates

Air Velocity (m/s)	Plasma Power (W)	Snail Fan Power (W)	Total Power (W)	Efficiency (η) $P_{\text{plasma}}/P_{\text{total}}$
4	100,8	0,42	101,22	0,99
8	96,454	1,14	97,594	0,98
12	92,4	2,4	94,8	0,97
16	88,83	4,8	93,63	0,94
20	85,02	8,75	93,77	0,93
24	81,25	13,32	94,57	0,85
28	77,52	19,74	97,26	0,80
32	73,83	28	101,83	0,72
36	70,4	38,7	109,1	0,64
40	63,5	50	113,5	0,55

Table 12 Numerical variation of the E/N (Td) value depending on different airflow velocity ratios

Air Velocity (m/s)	T_{gas} (K)	N (m^{-3})	E (V/m)	E/N (Td)
4	553	1.33×10^{25}	5.00×10^5	37.6
8	488	1.50×10^{25}	4.83×10^5	32.2
12	428	1.72×10^{25}	4.67×10^5	27.1
16	393	1.87×10^{25}	4.50×10^5	24.1
20	348	2.11×10^{25}	4.33×10^5	20.5
24	336	2.19×10^{25}	4.17×10^5	19.0
28	325	2.26×10^{25}	4.00×10^5	17.7
32	318	2.31×10^{25}	3.83×10^5	16.6
36	311	2.36×10^{25}	3.67×10^5	15.5
40	305	2.41×10^{25}	3.33×10^5	13.8

Fig. 18 Graphical representation of the change in E/N (Td) value depending on different airflow velocity ratios

physical indicators of the system's transition from a thermal to a non-thermal filamentous regime, and points to a change in the reaction mechanisms that have become dominant in plasma chemistry. The observed decrease in E/N with increasing airflow velocity supports the physical interpretation of the plasma regime transition discussed earlier. As E/N decreases, the average electron energy is reduced, which limits ionization processes and favors non-thermal excitation mechanisms. This behavior is consistent with the transition from a stable arc regime to a filamentary discharge structure, where localized and transient micro-discharges dominate.

To strengthen the reliability and physical interpretation of the experimental results obtained, the findings of this study were compared with studies reported in the existing literature. The results obtained in this study show significant consistency with the existing literature in terms of the optical emission characteristics and sensitivity to gas flow parameters of atmospheric gliding arc discharge plasmas.

It is known that the most dominant spectral emissions in atmospheric air plasmas generally consist of N_2 Second Positive System ($\text{C}^3\Pi_u \rightarrow \text{B}^3\Pi_g$) bands. Similarly, strong emissions observed around 337 nm have been reported as dominant in gliding arc plasmas in many studies. For example, in the gliding arc plasma study conducted by Marcinauskas

et al., it was shown that the 315 nm, 337 nm, 357 nm, and 380 nm bands are dominant in the spectrum of the air plasma [68]. These results are highly consistent with the optical emission spectra obtained in this study. The same study also indicated that the emissions observed around 391 nm correspond to the First Negative System (N_2^+) transitions of the nitrogen molecule ion and that these emissions are directly related to ionization processes. In this study, the lower intensity of the 391 nm emission compared to the 337 nm emission also suggests that ionization processes in the plasma are more limited than excitation processes. The effect of gas flow rate on plasma properties has been extensively studied in the literature. In gliding arc plasmas, increasing gas flow leads to elongation of the plasma column, thinning of the arc channel, and a transformation of the plasma into a filamentary character. A study by Czernichowski reported that reducing the gas flow rate increased the residence time of the gas in the plasma, resulting in a significant change in plasma chemistry [5]. Lower flow rates result in longer reaction times, increasing the formation of reactive species.

Similarly, studies by Fridman have shown that in gliding arc plasmas, gas flow rate is one of the most important parameters determining the plasma regime [6]. These studies have shown that at low gas flow rates, the plasma is in a more stable arc regime, while at high flow rates, the plasma transforms into a filamentary structure.

The plasma regime map obtained in this study also reveals a similar behavior. Determining plasma parameters using optical emission spectroscopy is a standard method used in many atmospheric plasma studies. In studies by Bibinov et al., it was shown that the electric field and electron energy distribution within the plasma can be determined by using the density ratios of N_2 bands [55].

The I_{391}/I_{337} density ratio used in this study similarly provides information about the ionization and excitation processes in the plasma. In terms of electron temperature, it has been reported in the literature that the electron temperature in gliding arc plasmas operating at atmospheric pressure is generally in the range of 1–5 eV. For example, a comprehensive study by Bruggeman and Leys showed that electron temperatures in atmospheric plasmas vary within this range [40]. The electron temperature values calculated in this study, in the range of 1.4–2.8 eV, are consistent with the range reported in the literature. From the perspective of plasma temperature equilibrium, atmospheric gliding arc discharges are known to generally exhibit a combination of thermal and non-thermal plasma characteristics. Studies by Fridman have shown that although the gas temperature of gliding arc plasmas remains at a level of a few hundred °C, the electron and vibrational temperatures can be much higher [6]. The large difference between the measured gas temperature (≈ 32 – 280 °C) and the calculated vibrational temperature (≈ 5000 – 16000 K) in this study indicates that the plasma is significantly far from thermal equilibrium. The effect of gas flow on plasma chemistry was also investigated, particularly in terms of NO_x formation. In a study by Marcinauskas et al., it was reported that increasing the air flow velocity decreased the NO concentration, and this was related to the residence time of the gas in the plasma [68].

This demonstrates that gas flow is a crucial parameter controlling plasma chemistry and reactive species production. Recent studies show that gliding arc plasmas have significant potential in energy conversion technologies and environmental applications. For example, a study by Wang et al. showed that gliding arc plasma can significantly reduce NO_x emissions in ammonia combustion systems [66]. These studies demonstrate that gliding arc plasmas can play a significant role in chemical transformation processes. Similarly, studies on rota-

tional gliding arc reactors have revealed that gas flow parameters have significant effects on plasma reactivity and energy efficiency [61].

When all these studies are evaluated together, it is seen that the results obtained in this study are consistent with plasma behaviors reported in the literature. However, this study offers a new perspective to the literature, especially in terms of analyzing the plasma regime, electron temperature, and optical emission density ratios together as a function of airflow velocity.

Overall Interpretation of the Plasma Behavior

When all experimental results are considered together, it is seen that airflow is a critical control parameter in atmospheric DC gliding arc plasmas. Airflow affects not only the geometry of the plasma channel but also the electrical properties, temperature distribution, and electron energy distribution of the plasma. Increased airflow cools the plasma channel, separates the plasma column into filaments, and alters the electron energy distribution. Therefore, airflow emerges as a parameter that directly affects the plasma regime and plasma chemistry.

Conclusions

This study experimentally investigates the plasma characteristics of a DC gliding arc discharge operating at atmospheric pressure as a function of airflow velocity. In a plasma system created using copper electrodes with divergence geometry, the electrical properties, gas temperature, optical emission characteristics, and plasma regime of the plasma were analyzed in detail by varying the airflow velocity. Electrical measurements showed that the discharge voltage decreased from approximately 3 kV to 2 kV with increasing airflow velocity. Furthermore, it was determined that the discharge current also gradually decreased from approximately 33.6 mA to approximately 31.7 mA depending on the airflow velocity. This can be explained by the convective cooling of the plasma channel by the airflow and the transformation of the plasma column into a finer filament structure. Gas temperature measurements showed that the airflow velocity had a significant cooling effect on the plasma thermodynamics. The measured gas temperature values decreased significantly from approximately 280 °C to 32 °C with increasing airflow velocity. This result indicates the presence of a strong convective cooling mechanism in the plasma channel. Optical emission spectroscopy analyses showed that the plasma consisted predominantly of nitrogen molecule bands. The strongest emission in the spectrum was observed at approximately 337 nm wavelength, corresponding to the Second Positive System transitions of nitrogen molecules.

In contrast, the First Negative System bands belonging to nitrogen molecule ions, observed around 391 nm, showed lower intensity. It was determined that the I_{391}/I_{337} intensity ratio, calculated as a function of air flow velocity, did not remain constant but rather showed significant variations depending on the flow velocity.

At low airflow velocities, this ratio shows higher values, while a significant decrease is observed at medium flow rates, and a fluctuating behavior reappears at higher flow rates. This indicates that the ionization and excitation processes in the plasma change with airflow, and the electron energy distribution is sensitive to flow conditions. The vibrational temperature values calculated using the Boltzmann method were found to be approximately in the

range of 5×10^3 K to 1.6×10^4 K. When these values are compared with the gas temperature, it reveals that the plasma has a distinct thermal instability (non-equilibrium plasma) character. When electrical measurements, optical emission analyses, and temperature measurements are evaluated together, it is seen that airflow velocity is an important parameter controlling the plasma structure. At low airflow velocities, the plasma is in a more stable arc regime, while with increasing airflow, the plasma channel breaks down and transforms into a filamentary discharge structure. This behavior is clearly demonstrated by the generated plasma regime map. One of the important results of this study is that it shows that atmospheric gliding arc plasma can be produced stably with a simple and low-cost DC power supply. Using readily available air as the discharge gas eliminates gas consumption compared to using expensive inert gases (Ar, He, Ne, etc.). This allows for the smooth execution of long plasma process times. Especially at high airflow rates, a cold plasma is obtained as the gas temperature, enabling controlled surface modifications and activations of temperature-sensitive materials. The ZVS-driven flyback transformer power supply offers a practical and economical plasma production method in a laboratory environment thanks to its compact structure, easy installation, easy transport without the need for extra equipment, and high voltage generation capacity. Furthermore, the system's operation at relatively low power levels limits energy consumption and provides an advantage in terms of energy efficiency in plasma production. In conclusion, this study demonstrates that airflow in atmospheric DC gliding arc plasmas can control fundamental plasma parameters such as plasma temperature, emission density, electron energy distribution, and discharge regime. The results obtained show that gliding arc plasmas can be used more controllably and efficiently in fields such as plasma chemistry, surface sterilization, hydrophilic-hydrophobic applications, gas conversion technologies (such as obtaining O_2 gas from CO_2 gas), as catalysts, environmental applications, and plasma-assisted chemical reactions. In particular, when the OES results are examined, the high amount of NO and OH formed suggests that their use is advantageous in the production of plasma active media (PAM) and in agriculture and fertilizer applications.

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Data Availability The authors declared that the data supporting the findings in this study are presented within the paper.

Declarations

Competing interests The authors declare no competing interests.

Ethical Approval The author declare that they comply with ethical standards. Experimental animals and humans were not used in this study. Therefore, an ethical permit was not required.

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